

VARUN GOPAL

PYTHON BACKEND ENGINEER | DJANGO | DJANGO REST FRAMEWORK | MYSQL

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Immediate joiner | Open to remote, in-office, and relocation opportunities

PROFESSIONAL SUMMARY

Python backend engineer with 4 years of experience building production REST APIs, optimizing MySQL workloads, and integrating third-party services in healthcare technology. Hands-on expertise in Django, Django REST Framework, SQL performance tuning, stored procedures, JWT authentication, multi-tenant access control, asynchronous data synchronization, and AWS services. Experienced in delivering secure, observable backend systems and collaborating across Agile, DevOps, and frontend teams.

TECHNICAL SKILLS

Languages & Backend: Python, Django, Django REST Framework, Flask, JavaScript (basic), REST API design, microservices

Databases & Storage: MySQL, SQL, stored procedures, functions, indexing, EXPLAIN, query optimization, ChromaDB, AWS S3

Cloud, DevOps & Tools: AWS Lambda, AWS CloudWatch, Docker, Jenkins CI/CD, Git, Linux, cron jobs, blue-green deployment, Jira

Security, Testing & API Tools: JWT authentication, role-based access control, multi-tenant architecture, Pytest, Postman, Swagger

AI & Agentic Systems: Google Agent Development Kit (ADK), Gemini API, FastMCP, retrieval-augmented generation (RAG), Pydantic, custom function tools, multi-agent orchestration

PROFESSIONAL EXPERIENCE

Senior Software Engineer

June 2022 - July 2026

Equipo Health Inc. | Trivandrum, Kerala

- Built and maintained Django REST Framework APIs for patient records, referrals, e-consultations, and clinical reporting within a microservices architecture; supported workflows operating across millions of database rows and engineered endpoints toward sub-100 ms response targets.
- Led MySQL performance tuning for high-traffic endpoints using EXPLAIN analysis, composite and covering indexes, and query rewrites; eliminated full table scans and filesorts and moved complex reporting aggregations into stored procedures to reduce parsing and application-to-database round-trips.
- Implemented JWT middleware and role-based permission checks with local token verification, avoiding per-request database lookups. Enforced multi-tenant isolation by scoping patient and document queries to the group or facility encoded in the authenticated user's token.
- Developed secure clinical document workflows with file validation, private AWS S3 storage, MySQL metadata, and short-lived pre-signed URLs protected by patient-level authorization checks.
- Designed asynchronous, queue-based synchronization between the core platform and an external care-management system, including event triggers, idempotent upserts keyed by unique event ID, duplicate prevention, and conflict-resolution rules.
- Built scheduling and eligibility logic for appointment reminders, medication nudges, and follow-ups delivered through the Telnyx API via SMS, IVR, and email; added timezone-aware delivery windows and database-level status locks to prevent duplicate sends.

- Delivered APIs for a React-based operations control panel covering facility configuration, referral routing, and user permissions; added server-side pagination, filtering, and select_related optimizations to prevent N+1 queries.
- Improved production traceability with Django middleware that generated request IDs, emitted structured JSON logs, and exposed trace IDs in response headers for AWS CloudWatch Logs Insights. Also built scheduled Python batch jobs and contributed to Docker, Jenkins CI/CD, and blue-green deployment workflows.

PROJECTS

CivicPulse AI - Multi-Agent Civic Grievance & Escalation System

Self-directed project | Google ADK, FastMCP, Django REST Framework, MySQL, ChromaDB, Gemini API

- Designed a municipal grievance platform with a Google ADK coordinator agent that routes citizen requests to specialized Triage, Knowledge, and Escalation agents based on intent and permitted tool access.
- Built a FastMCP server over streamable HTTP with deterministic, raw-SQL-backed tools for category lookup, complaint filing, service-level agreement (SLA) breach checks, and geospatial duplicate detection using bounding-box filtering and a custom haversine calculation.
- Modeled configurable SLA rules by category and priority, including fallback behavior, to drive deadline computation consistently outside the language model.
- Implemented a retrieval-augmented generation pipeline over a grievance charter using ChromaDB and Gemini embeddings with separate retrieval-document and retrieval-query task types for grounded policy responses.
- Added before-model prompt filtering and before-tool audit logging, then bridged synchronous Django REST Framework views to the asynchronous ADK Runner through a dedicated background asyncio event loop.
- Used MySQL-backed session persistence and fast-path ORM endpoints for status and statistics polling that bypass the language model.

ADDITIONAL TRAINING

Google Skills Coursework - Agentic AI & Tool Integration

Google ADK, Gemini API, Pydantic, Python

- Built agents with structured Pydantic outputs, session-state templating, custom function tools, Google Search grounding, and code execution tools.
- Configured agents through programmatic Python and declarative YAML aligned with Google ADK schemas.

EDUCATION

Bachelor of Engineering (B.E.) in Computer Science and Engineering

SNS College of Engineering, Anna University | Coimbatore, Tamil Nadu | 2014 - 2018

CERTIFICATIONS

- Engineer AI Agents with Agent Development Kit (ADK) - Google Cloud
- Create Your First Gemini Enterprise Application - Google Cloud
- SQL (Advanced) - HackerRank
- API Security Fundamentals - APIsec University
- Problem Solving (Intermediate) - HackerRank
- Data Processing and Visualization - Accenture via SSC NASSCOM

LANGUAGES

English (Professional Working Proficiency) | Malayalam (Native) | Tamil (Professional Working Proficiency)